SE4a_u1_01k_001_1

Which of these items is the most similar to the structure of the Earth?

A. watermelon
B. apple
C. boiled egg
D. table tennis ball

( )

[--Ans: C--]

SE4a_u1_01k_002_1

What is the Earth's crust mainly made up of?

A. magma
B. rocks
C. soil
D. sand

( )

[--Ans: B--]

SE4a_u1_01k_003_1

What is the centre of the Earth?

A. rocks
B. Earth's crust
C. Earth's mantle
D. Earth's core

( )

[--Ans: D--]

SE4a_u1_01k_004_2

Which of these descriptions about the structure of the Earth is correct?

A. The Earth's crust is the centre of the Earth.
B. The Earth's core is the outermost layer of the Earth.
C. The Earth's mantle is mainly made up of rocks.
D. The Earth's core is mainly made up of metals. The temperature is extremely high.

( )

[--Ans: D--]

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SE4a_u1_01k_007_1

What natural phenomena may plate movement and collision cause? (There can be more than one answer.)

A. heavy rain
B. lightnings
C. earthquakes
D. volcanic eruptions

( )

[--Ans: C, D--]

SE4a_u1_01k_008_1

What type of rock is basalt?

A. igneous rock
B. sedimentary rock
C. metamorphic rock
D. none of the above

( )

[--Ans: A--]

SE4a_u1_01k_009_1

What characteristic of the soil can you observe by using a sieve to sieve the soil?

A. whether it contains water
B. whether it contains air
C. the particle size
D. the weight

( )

[--Ans: C--]

SE4a_u1_01k_010_1

Which type of soil has the smallest particles: sandy soil, loam or clay soil?

A. sandy soil
B. loam
C. clay soil
D. not sure

( )

[--Ans: C--]

SE4a_u1_01k_011_1

Which type of soil has the best drainage capacity: sandy soil, loam or clay soil?

A. sandy soil
B. loam
C. clay soil
D. not sure

( )

[--Ans: A--]

SE4a_u1_01k_012_1

Imagine a torch represents the Sun and shines on a globe. When Hong Kong on the globe is facing and shone by the Sun, what time of the day is it in Hong Kong?

A. noon
B. evening
C. midnight
D. not sure

( )

[--Ans: A--]

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SE4a_u1_01k_013_2

Which of these descriptions about the Earth is correct? (There can be more than one answer.)

A. It takes around 24 hours for the Earth to rotate once.
B. The Sun moves around the Earth.
C. The cycle of day and night is caused by the rotation of the Earth.
D. The imaginary axis of the Earth is tilted.

( )

[--Ans: A, C, D--]

SE4a_u1_01k_014_1

Which of these descriptions about the Earth's rotation is correct? (There can be more than one answer.)

A. The Earth keeps rotating around the axis every day.
B. The Earth's axis is real. It passes through the North Pole and the South Pole.
C. We see sunrise and sunset because the Earth rotates.
D. It takes around 365 days for the Earth to rotate once.

( )

[--Ans: A, C--]

SE4a_u1_01k_016_2

How long does it take for the Earth to rotate once?

A. 12 hours
B. 24 hours
C. 1 month
D. 1 year

( )

[--Ans: B--]

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SE4a_u1_01k_017_1

What changes in the length of a shadow can you notice from morning to noon?

A. The shadow becomes longer.
B. The shadow becomes shorter.
C. The shadow becomes longer first and then shorter.
D. The shadow becomes shorter first and then longer.

( )

[--Ans: B--]

SE4a_u1_01k_019_1

How long does it take for the Earth to complete one revolution?

A. 1 hour
B. 1 month
C. 1 day
D. 1 year

( )

[--Ans: D--]

SE4a_u1_01k_020_1

When the northern hemisphere receives overhead sunlight, what season is it in Hong Kong?

A. spring
B. summer
C. autumn
D. winter

( )

[--Ans: B--]

SE4a_u1_01k_021_1

When the northern hemisphere receives slanted sunlight, what season is it in Hong Kong?

A. spring
B. summer
C. autumn
D. winter

( )

[--Ans: D--]

SE4a_u1_01k_022_1

When the northern hemisphere receives overhead sunlight, what season is it in Melbourne in the southern hemisphere?

A. spring
B. summer
C. autumn
D. winter

( )

[--Ans: D--]

SE4a_u1_01k_023_1

Which of these descriptions about the revolution of the Earth is correct?

A. The Earth moves around the Sun.
B. It takes around 24 hours for the Earth to complete one full revolution.
C. When the Earth moves to different locations during a revolution, the temperature in the southern hemisphere remains the same.
D. When the Earth is moving around the Sun, the southern hemisphere and the northern hemisphere receive sunlight at the same angle.

( )

[--Ans: A--]

SE4a_u1_01k

What can you use to describe the weather of a place? (There can be more than one answer.)

A. temperature
B. relative humidity
C. rainfall
D. climate

( )

[--Ans: A, B, C--]

SE4a_u1_01k_026_1

When it is summer in Hong Kong, which of these places is in summer too? (There can be more than one answer.)

A. Beijing
B. Shanghai
C. Singapore
D. Canberra

( )

[--Ans: A, B, C--]

SE4a_u1_01k_027_1

When it is summer in Hong Kong, which of these places is in winter?

A. Beijing
B. Shanghai
C. Singapore
D. Canberra

( )

[--Ans: D--]

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SE4a_u1_01k_028_1

Which of these places do not have four distinct seasons?

A. Beijing
B. Shanghai
C. Singapore
D. Canberra

( )

[--Ans: C--]

SE4a_u1_01k_030_1

Which of these activities cause the amount of carbon dioxide in the Earth's atmosphere to increase sharply? (There can be more than one answer.)

A. deforestation
B. factories give out waste gas
C. cars give out waste gas
D. planting a lot of trees

( )

[--Ans: A, B, C--]

SE4a_u1_01k_031_1

Which of these is not an effect of global warming?

A. melting glaciers
B. rising sea levels
C. desertification
D. excessive use of fuels such as coal and oil

( )

[--Ans: D--]

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SE4a_u1_01a_001_1

What should we do to slow down global warming? (There can be more than one answer.)

A. do not waste food
B. use single-use utensils
C. turn off electrical appliances that are not in use
D. only buy what we need

( )

[--Ans: A, C, D--]

SE4a_u1_01a_002_2

What should we avoid doing to slow down global warming? (There can be more than one answer.)

A. save electricity
B. waste food
C. use single-use utensils
D. take public transport

( )

[--Ans: B, C--]

SE4a_u1_01a_003_1

What should we avoid doing to reduce carbon footprints? (There can be more than one answer.)

A. use single-use utensils
B. use electrical appliances that save energy
C. buy things that we do not practically need
D. travel in private cars and taxis

( )

[--Ans: A, C, D--]

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SE4a_u1_02k_002_2

The structure of the Earth is made up of three layers. From the inside to the outside, they are the _____, the _____ and the _____.

[--Ans: core, mantle, crust--]

SE4a_u1_02k_003_1

The centre of the Earth is the _____. The temperature is extremely high.

[--Ans: core --]

SE4a_u1_02k_005_1

We live on the _____. It is the outermost layer of the Earth.

[--Ans: crust--]

SE4a_u1_02k_006_1

The crust is mainly made up of _____.

[--Ans: rocks--]

SE4a_u1_02k_007_1

The mantle contains hot flowing _____.

[--Ans: magma--]

SE4a_u1_02k_009_1

The crust is the outermost layer of the Earth. It is divided into _____ of different sizes.

[--Ans: plates--]

SE4a_u1_02k_010_1

The movement and collision of plates may cause _____ and _____.

[--Ans: earthquakes, volcanic eruptions (Flexible order.)--]

SE4a_u1_02k_013_1

Some rocks break into pieces after many years of exposure to wind and rain, forming _____.

[--Ans: soil--]

SE4a_u1_02k_014_1

Rocks can be divided into three types according to how they are formed: _____, _____ and _____.

[--Ans: igneous rock, sedimentary rock, metamorphic rock (Flexible order.)--]

SE4a_u1_02k_015_1

Rocks formed when magma cools and hardens are called _____.

[--Ans: igneous rocks--]

SE4a_u1_02k_016_1

Rocks formed when different types of rocks change their structures under high pressure and high temperature are called _____.

[--Ans: metamorphic rocks--]

SE4a_u1_02k_017_1

Rocks formed by the sedimentation of rock pieces and soil transported down by water are called _____.

[--Ans: sedimentary rocks--]

SE4a_u1_02k_018_1

The components of soil include animal and plant matter, _____, _____ and _____.

[--Ans: soil particles, water, air (Flexible order.)--]

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SE4a_u1_02k_019_1

According to the size of particles from small to large, we can divide soil into _____, _____ and _____.

[--Ans: clay soil, loam, sandy soil--]

SE4a_u1_02k_020_1

Among sandy soil, loam and clay soil, _____ has the best drainage capacity, _____ has the worst drainage capacity and _____ has moderate drainage capacity.

[--Ans: sandy soil, clay soil, loam--]

SE4a_u1_02k_021_1

The Earth rotates on an imaginary _____ that passes through the North Pole and South Pole every day.

[--Ans: axis--]

SE4a_u1_02k_022_1

The Earth _____ continuously. This causes the cycle of day and night.

[--Ans: rotates--]

SE4a_u1_02k_023_1

It takes around _____ hours for the Earth to rotate once.

[--Ans: twenty-four / 24--]

SE4a_u1_02k_024_1

At different times of the day, the _____ and _____ of the shadows formed in the sunlight change.

[--Ans: length, direction (Flexible order.)--]

SE4a_u1_02k_025_1

In the past, people knew the time by observing the _____ of objects. The device for observing shadows is called a sundial

[--Ans: shadows--]

SE4a_u1_02k_027_1

The Earth moves around the _____. This is called revolution.

[--Ans: Sun --]

SE4a_u1_02k_028_1

The Earth moves around the Sun. This is called _____.

[--Ans: revolution --]

SE4a_u1_02k_029_1

It takes _____ for the Earth to revolve around the Sun once.

[--Ans: one year / about 365 days--]

SE4a_u1_02k_030_1

As the Earth's axis is _____, the angle of sunlight hitting the Earth changes on different days. This causes the changes of _____.

[--Ans: tilted, seasons--]

SE4a_u1_02k_031_1

When the southern hemisphere receives overhead sunlight, it is _____ in the southern hemisphere and it is _____ in the northern hemisphere.

[--Ans: summer, winter--]

SE4a_u1_02k_032_1

We can know the coldness and the hotness of the weather by the _____ announced by the Hong Kong Observatory.

[--Ans: temperature--]

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SE4a_u1_02k_033_1

We can know the dryness and the wetness of the weather by the _____ announced by the Hong Kong Observatory.

[--Ans: relative humidity--]

SE4a_u1_02k_034_1

Humans use large amounts of fuels like _____ and _____. This releases large amounts of greenhouse gases.

[--Ans: gas, oil (Flexible order.)--]

SE4a_u1_02k_035_1

Humans use large amounts of fuels like gas and oil. This releases large amounts of _____ and causes global warming.

[--Ans: carbon dioxide / greenhouse gases--]

SE4a_u1_02k_036_1

Carbon dioxide is a greenhouse gas. It _____ the heat from releasing from the Earth's surface and causes a rise in the Earth's temperature.

[--Ans: blocks--]

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SE4a_u1_03k_001_1

rocks	crust	mantle	core

1. The _____ is the centre of the Earth. It is mainly made up of metals and has a very high temperature.
2. The outermost layer of the Earth is mainly made up of _____.
3. The _____ contains hot flowing magma.
4. Human activities take place mainly on the _____.

[--Ans: 1. core; 2. rocks; 3. mantle; 4. crust--]

SE4a_u1_03k_002_1

core	magma	earthquakes	plates	mantle

1. The crust is divided into _____ of different sizes.
2. Magma in the mantle flows so _____ move slowly too.
3. Plate movement and collision may cause volcanic eruptions or _____.
4. During a volcanic eruption, _____ gushes out of the volcano.

[--Ans: 1. plates; 2. mantle; 3. earthquakes; 4. magma--]

SE4a_u1_03k_003_1

soil	rocks	igneous rocks
sandstones	sedimentary rocks	metamorphic rocks

1. The crust is mainly made up of _____.
2. _____ are rocks that are formed when magma cools and hardens.
3. _____ are rocks that are formed when different types of rocks change their structures under high pressure and high temperature.
4. _____ are rocks that are formed by the sedimentation of rock pieces and soil transported down by water.

[--Ans: 1. rocks; 2. Igneous rocks; 3. Metamorphic rocks; 4. Sedimentary rocks--]

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SE4a_u1_03k_005_1

east	direction	longer
west	length	shorter

1. From morning to noon, the shadow gets _____.
2. After noon, the shadow gets _____.
3. In a day, the shadow changes from pointing to the _____ in the morning to pointing to the _____ in the afternoon.
4. The _____ and _____ of a shadow are different at different times of the day.

[--Answer: 1. shorter; 2. longer; 3. west, east; 4. length, direction (Flexible order.)--]

SE4a_u1_03k_007_1

weather	climate	relative humidity
temperature	northern hemisphere	southern hemisphere

1. The Hong Kong Observatory uses _____ to describe the coldness and the hotness of the weather.
2. The Hong Kong Observatory uses _____ to describe the dryness and the wetness of the weather.
3. Hong Kong is in the _____. Canberra is in the _____.
4. The average rainfall in Hong Kong is higher in summer. The _____ is quite humid.

[--Ans: 1. temperature; 2. relative humidity; 3. northern hemisphere, southern hemisphere; 4. climate--]

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Fill in the blanks with word choices
(Knowledge & understanding)

SE4a_u1_03k_008_1

high waves	carbon dioxide	rising
falling	storm surge	very hot

1. Humans use large amounts of fuels like gas and oil. This releases large amounts of _____ and causes global warming.
2. Global warming causes _____ sea levels and desertification.
3. Rising sea levels increase the risk of coastal areas being hit by _____ and _____.
4. Due to global warming, the number of days with _____ weather is increasing. We should take protective measures against the heat.

[--Ans: 1. carbon dioxide; 2. rising; 3. storm surge, high waves (Flexible order.); 4. very hot--]

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Unit 1 About the Earth

Fill in the blanks with word choices
(Skills & processes)

SE4a_u1_03s_001_1

fingernail	magnifying glass	colours	sandstones
weight	granite	hardness	

1. When we observe rock samples, we can first observe their _____ and size of particles.
2. We can use a _____ to observe the size of the particles in a rock sample.
3. We can scratch rocks with a stainless steel key and a _____ to see if there are any scratches and compare the _____ of the rocks.

[--Ans: 1. colours; 2. magnifying glass; 3. fingernail, hardness--]

SE4a_u1_03s_002_1

particles	sieve	magnifying glass	measuring cup
bubbles	drier	digital scale	

1. We can use a _____ to observe the soil and look for any animal and plant matter (e.g. dead leaves, insects) in the soil.
2. Set the weight of the _____ to zero. Rub the soil and use the digital scale to weigh it. Then, put the soil under the Sun or in a place with air flow. After a week, measure its weight again. Rub the soil and observe if it is moister or _____.
3. Use a _____ to sieve the dry soil. Observe the size of the _____.
4. Put some soil into a _____ filled with water. Stir it with a plastic rod. Check if there are any _____ on the water surface.

[--Ans: 1. magnifying glass; 2. digital scale, drier; 3. sieve, particles; 4. measuring cup, bubbles--]

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SE4a_u1_03s_003_1

different	same height	sandy soil	loam
the same	drainage capacity	clay soil	

1. To test the _____ of different types of soil, cut three plastic bottles in half with a knife.
2. Fix a piece of gauze to the mouth of each bottle with a rubber band. Turn them upside down and put them on the cut bottles. Then, put the same amount of soil sample on the gauze so that all the samples have the _____.
3. Put _____ amount of water into the three plastic bottles. After five minutes, pour the water into measuring cups and measure the amount of water.
4. The _____ releases the most water and the _____ releases the least.

[--Ans: 1. drainage capacity; 2. same height; 3. the same; 4. sandy soil, clay soil--]

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SE4a_u1_04k_001_1

The outermost layer of the Earth is the crust. ( )
[--Ans: ✓--]

SE4a_u1_04k_002_1

The mantle contains hot flowing magma. ( )
[--Ans: ✓--]

SE4a_u1_04k_003_1

The centre of the Earth is the mantle, which has a very high temperature. ( )
[--Ans: ×--]

SE4a_u1_04k_004_1

The centre of the Earth is the crust. It is mainly made up of rocks. The temperature is very high. ( )
[--Ans: ✓--]

SE4a_u1_04k_005_1

The crust is mainly made up of a complete piece of rock. ( )
[--Ans: × --]

SE4a_u1_04k_006_1

Plates move slowly when the magma in the mantle flows. ( )
[--Ans: ✓--]

SE4a_u1_04k_007_1

Earthquake is a natural phenomenon caused by plate movement and collision. ( )
[--Ans: ✓--]

SE4a_u1_04k_008_1

Shale is a kind of igneous rock. ( )
[--Ans: ×--]

SE4a_u1_04k_009_2

Both marbles and gneiss are sedimentary rocks. ( )
[--Ans: ×--]

SE4a_u1_04k_011_1

Different kinds of soil have different drainage capacities. Sandy soil has the weakest drainage capacity, while clay soil has the strongest drainage capacity. ( )
[--Ans: ×--]

SE4a_u1_04k_012_1

The Earth rotates on a real axis every day. ( )
[--Ans: ×--]

SE4a_u1_04k_013_1

We can see the Sun rising and setting every day because of the Earth's rotation. ( )
[--Ans: ✓ --]

SE4a_u1_04k_014_1

During the day, the length of a shadow changes at different times. However, the direction of the shadow does not change. ( )
[--Ans: ×--]

SE4a_u1_04k_015_1

A sundial is a device that people in the past used to tell the time by observing the shadow. ( )
[--Ans: ✓--]

SE4a_u1_04k_016_1

The Earth moves around the Moon. This is called revolution. ( )
[--Ans: ×--]

SE4a_u1_04k_017_1

Day and night repeats regularly because the Sun moves around the Earth. ( )
[--Ans: ×--]

SE4a_u1_04k_018_1

When the northern hemisphere receives overhead sunlight, it is winter in the northern hemisphere and summer in the southern hemisphere. ( )

[--Ans: ×--]

SE4a_u1_04k_019_1

There is a leap year once in every 5 years. ( )

[--Ans: ×--]

SE4a_u1_04k_021_1

Climate is the average weather condition of a place over a longer period of time. ( )

[--Ans: ✓--]

SE4a_u1_04k_022_2

Singapore and Beijing have a similar climate because they are both in the northern hemisphere. ( )

[--Ans: ×--]

SE4a_u1_04k_024_1

Melting of land ice causes sea levels to rise. ( )

[--Ans: ✓--]

SE4a_u1_04k_025_1

Carbon dioxide blocks the heat from releasing from the Earth's surface and causes global warming. ( )

[--Ans: ✓--]

SE4a_u1_04k_026_1

Carbon footprints are the amount of greenhouse gases produced in some human activities. ( )

[--Ans: ✓--]

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SE4a_u1_04a_001_1

Rocks and soil are precious resources. We should cherish and protect them. ( )

[--Ans: ✓--]

SE4a_u1_04a_002_1

Due to global warming, the number of days with very hot weather is increasing. We should pay attention to the Very Hot Weather Warning and take protective measures against the heat. ( )

[--Ans: ✓--]

SE4a_u1_04a_004_1

Wilson thinks that one's influence is limited so he does not reduce his carbon footprints in daily life. ( )

[--Ans: ×--]

SE4a_u1_05k_001_1

The internal structure of the Earth from the inside to the outside is the crust, the mantle and the core.

( )

[--Ans: ×, the core, the mantle, the crust--]

SE4a_u1_05k_003_1

The crust is not a complete piece of rock. It is made up of plates of different sizes.

( )

[--Ans: ✓--]

SE4a_u1_05k_005_1

Soil is formed when rocks break into small pieces after being exposed to wind and rain for many years.

( )

[--Ans: ✓--]

SE4a_u1_05k_007_1

Sedimentary rocks are rocks formed when different types of rocks change their structures under high pressure and high temperature.

( )

[--Ans: ×, Metamorphic rocks--]

SE4a_u1_05k_008_1

Different types of soil have different drainage capacities. Loam has the highest drainage capacity.

( )

[--Ans: ×, Sandy soil--]

SE4a_u1_05k_009_1

Day and night keep repeating. This is because the Earth rotates non-stop.

( )

[--Ans: ✓--]

SE4a_u1_05k_010_1

We can see the Sun rises and sets every day. This is because the Earth's revolution.

( )

[--Ans: ×, rotation--]

SE4a_u1_05k_011_1

Every day, the Earth rotates around a real axis that passes through the North Pole and the South Pole.

( )

[--Ans: ×, an imaginary--]

SE4a_u1_05k_012_1

During the day, the length and direction of a shadow change at different times.

( )

[--Ans: ✓--]

SE4a_u1_05k_013_1

When the northern hemisphere receives overhead sunlight, it is winter in the northern hemisphere.

( )

[--Ans: ×, summer--]

SE4a_u1_05k_015_1

Climate is the average weather condition over a long period of time in the past.

( )

[--Ans: ✓ --]

SE4a_u1_05k_017_1

Humans use large amounts of fuels like gas and oil. This releases large amounts of greenhouse gases and causes global warming.

( )

[--Ans: ✓--]

SE4a_u1_07k_001_1

Match the layers of the Earth with their features.

1. crust
2. mantle
3. core

A. It contains hot flowing magma.
B. It is the centre of the Earth. It is mainly made up of metals. The temperature is very high.
C. It is made up of different types of gases. It surrounds the Earth.
D. It is the outermost layer of the Earth. It is mainly made up of rocks and is divided into many plates.

1. _____ 2. _____ 3. _____

[--Ans: 1. D; 2. A; 3. B--]

SE4a_u1_07k_002_1

What are the characteristics of the Earth's layers? (There can be more than one answer.)

1. characteristics of the crust
2. characteristics of the mantle
3. characteristics of the core

A. mainly made up of rocks
B. mainly made up of metals
C. contains hot flowing magma
D. the centre of the Earth
E. the outermost layer of the Earth

1. _____ 2. _____
3. _____

[--Ans: 1. A, E; 2. C; 3. B, D--]

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SE4a_u1_07k_003_1

Match the parts of the crust with their characteristics. (There can be more than one answer.)

1. plates
2. rocks
3. soil

A. formed when magma gushing out from volcanoes cools and hardens
B. its components are water, air, animal and plant matter and soil particles
C. slowly move with the flow of magma in the mantle
D. their movements and collisions can cause earthquakes or volcanic eruptions

1. _____ 2. _____
3. _____

[--Ans: 1. C, D; 2. A; 3. B--]

SE4a_u1_07k_004_1

Match the types of rocks with their formation processes.

1. igneous rocks
2. sedimentary rocks
3. metamorphic rocks

A. They are formed when different types of rocks change their structures under high pressure and high temperature.
B. They are formed when magma cools and hardens.
C. They are formed by the sedimentation of rock pieces and soil transported down by water.

1. _____ 2. _____ 3. _____

[--Ans: 1. B; 2. C; 3. A--]

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SE4a_u1_07k_006_2

Match the common rocks with their characteristics.

1. granite
2. sandstone
3. marble

A. usually with patches of pink, grey and other colours
B. usually orange and grey, and looks like different sizes of sand glued together
C. usually grey or white with beautiful stripes

1. _____ 2. _____
3. _____

[--Ans: 1. A; 2. B; 3. C--]

SE4a_u1_07k_007_1

Match the different types of soil with their properties. (There can be more than one answer.)

1. sandy soil
2. loam
3. clay soil

A. Its particles are the largest among the three types of soil.
B. It keeps nutrients the best among the three types of soil.
C. It is suitable for growing most kinds of plants.
D. It is suitable for growing cacti.

1. _____ 2. _____
3. _____

[--Ans: 1. A, D; 2. C; 3. B--]

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SE4a_u1_07k_008_1

What natural phenomena do the rotation and revolution of the Earth cause? (There can be more than one answer.)

1. Earth's rotation
2. Earth's revolution

A. the day-night cycle
B. change of seasons
C. the change in the length and direction of a shadow at different times of the day
D. the change in the angle of sunlight that the northern and southern hemispheres receive

1. _____ 2. _____

[--Ans: 1. A, C; 2. B, D--]

SE4a_u1_07k_012_1

Compare Beijing, Singapore and Canberra with Hong Kong. How are their geographic locations and climate? (There can be more than one answer.)

1. Beijing
2. Singapore
3. Canberra

A. It is in a different hemisphere from Hong Kong.
B. It is nearer to the equator than Hong Kong.
C. It has similar annual temperature changes with Hong Kong.
D. It has a distinct rainy season and dry season like Hong Kong.

1. _____ 2. _____
3. _____

[--Ans: 1. C, D; 2. B; 3. A--]

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SE4a_u1_07k_013_1

What do Hong Kong and Singapore have in common? What do they not have in common? (There can be more than one answer.)

1. what Hong Kong and Singapore have in common	A. located in the northern hemisphere
2. what Hong Kong and Singapore do not have in common	B. located near the equator
	C. has four distinct seasons
	D. more rainfall in summer
	E. does not snow in winter

1. _____ 2. _____

[--Ans: 1. A, E; 2. B, C, D--]

SE4a_u1_07k_014_2

What are the causes of rising sea levels? What are the effects of rising sea levels? (There can be more than one answer.)

1. causes of rising sea levels	A. more frequent storm surges
2. effects of rising sea levels	B. global warming
	C. low-lying areas being flooded
	D. melting of land ice

1. _____ 2. _____

[--Ans: 1. B, D; 2. A, C--]

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SE4a_u1_07s_001_1

When we simulate the rotation of the Earth, what do the different items or phenomena represent?

1. torch	A. the Earth
2. globe	B. it is night in Hong Kong
3. the paper puppet fixed at the location of Hong Kong is bright	C. it is day in Hong Kong
4. the paper puppet fixed at the location of Hong Kong is dark	D. the Sun

1. _____ 2. _____ 3. _____ 4. _____

[--Ans: 1. D; 2. A; 3. C; 4. B--]

SE4a_u1_07s_002_1

In an experiment that simulates the melting of sea ice and land ice, what do the different items or phenomena represent?

1. water	A. rise in sea level
2. modelling clay	B. seawater
3. ice in the water	C. land
4. ice on the modelling clay	D. sea ice
5. water level in the box rises	E. land ice

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

[--Ans: 1. B; 2. C; 3. D; 4. E; 5. A--]

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SE4a_u1_07a_001_1

Match the behaviours towards rock and soil resources with the effects.

1. take away rocks from Geopark	A. cherishes soil resources
2. litter on soil	B. damages the park
3. put the soil back into the flower bed after observation	C. pollutes the soil
	D. loses soil

1. _____ 2. _____ 3. _____

[--Ans: 1. B; 2. C; 3. A--]

SE4a_u1_07a_002_1

Match the human activities with the relationship to global warming.

1. recycle food waste	A. increases personal carbon footprints
2. drive private cars	B. reduces carbon footprints from food
3. factories giving out waste gas	C. absorbs carbon dioxide
4. plant trees	D. releases large amounts of carbon footprints in the production process

1. _____ 2. _____

3. _____ 4. _____

[--Ans: 1. B; 2. A; 3. D; 4. C--]

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SE4a_u1_08k_001_1

Put the layers of the Earth in order from the outside to the inside.

A. Mantle
B. Core
C. Crust

(　　) → (　　) → (　　)

[--Ans: C → A → B--]

SE4a_u1_08k_002_1

Put the formation processes of rocks in the correct order.

A. Magma gushes out of a volcano.
B. It causes a volcanic eruption.
C. Plates move and collide with one another.
D. Rocks are formed when magma cools and hardens.

(　　) → (　　) → (　　) → (　　)

[--Ans: C → B → A → D--]

SE4a_u1_08s_001_2

Put the steps of observing and recording shadows in the correct order.

A. Every hour, mark the position of the top of the shadow on the paper and record the time.
B. Measure the length of the line with a ruler. Record the results.
C. After recording, draw a line between the top of each shadow and the point where the pencil is stood.
D. Find an open area. Use a compass to find the north. Align the north on the paper with the north on the compass.

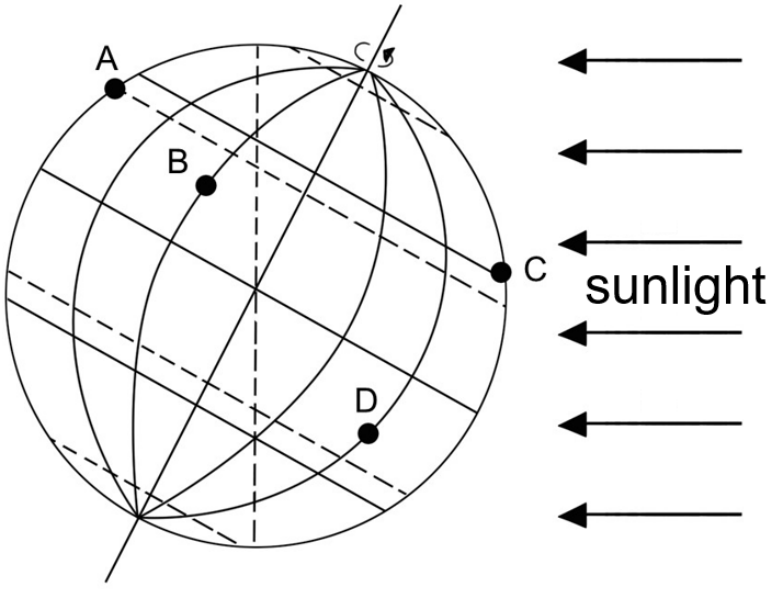
( ) → ( ) → ( ) → ( )

[--Ans: D → A → C → B--]

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SE4a_u1_09k_001_1

Look at the picture and think about the questions. Circle the answers or write the answers on the ____.



1. A, B, C and D are in four different time zones on the Earth. A, B and C are in the _____ while D is in the _____.

2. According to the picture, it is (daytime / nighttime) at C while it is (daytime / nighttime) at A.

3. According to the picture, it is (summer / winter) in A, B and C while it is (summer / winter) at D.

[--Ans: 1. northern hemisphere, southern hemisphere; 2. daytime, nighttime; 3. summer, winter--]

SE4a_u1_09s_001_1

Look at the picture and think about the questions. Write the correct answers in the ( ).

When Jenny is doing an experiment, she needs to explore this rock. She observes it carefully and finds that it is pure white with beautiful patterns. The particles are coarse, shiny and beautiful.



1. Jenny scratches the rock with her fingernail and a stainless steel key. Only the key can leave marks on the rock. The hardness of the rock should be ( ).
   A. lower than both her fingernail and the key
   B. higher than her fingernail but lower than the key
   C. higher than both her fingernail and the key

2. Judging from the characteristics of the rock, it is ( ).
   A. granite B. marble C. sandstone

3. We often use this kind of rock ( ).
   A. make sculptures
   B. cut metals
   C. to extract minerals and metals

[--Ans: 1. B; 2. B; 3. A--]

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SE4a_u1_09s_004_1

Look at the picture and think about the questions. Write the correct answers in the ( ).

Fred takes some soil from a flower bed in school and puts it in a plastic bag. He brings it back to the classroom, spreads it on a piece of paper to dry, and breaks it into small pieces. He then uses a magnifying glass to observe the soil. Finally, he puts the soil into water to study its composition.



1. After drying the soil on the paper, its weight will ( ).
   A. decrease B. increase C. have no change

2. When Fred observes the water with a magnifying glass, he may see ( ).
   A. a iron nail B. a living earthworm C. some tree leaves

3. Fred stirs the soil in the water with a plastic rod. He sees bubbles coming out. This shows that there is ( ) in the soil.
   A. water B. air C. sand

[--Ans: 1. A; 2. C; 3. B--]

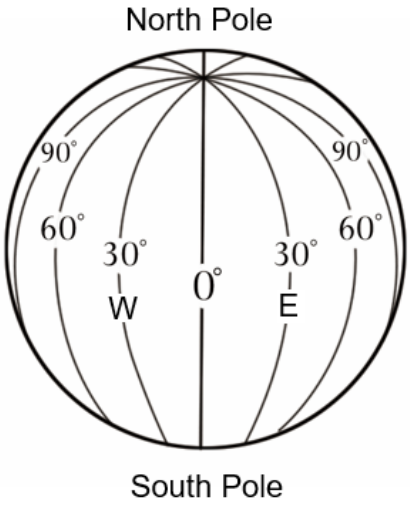
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SE4a_u1_09s_005_2

Look at the picture and think about the questions. Write the correct answers in the ( ).

Fred and Jenny are curious about the time difference between different places. They found out that because the Earth rotates from west to east, the time of sunrise is different in different places. Places in the east see the sunrise earlier than places in the west.

Look at the globe on the right. The lines connecting the North Pole and the South Pole are called meridians. They are imaginary lines for humans to measure locations on the Earth. The meridian that passes through the Royal Greenwich Observatory in London, the United Kingdom, is set as 0° longitude. It is also called the prime meridian.



The meridians east of the prime meridian are called east longitudes, while the meridians west of the prime meridian are called west longitudes. For every 15° difference in longitude, the time of sunrise differs by one hour. For every 30° difference, the time of sunrise differs by two hours.

1. The main cause of the time difference between different places on the Earth is ( ).
   A. the Earth's revolution B. the Earth's rotation
   C. the tilt of the Earth's axis

2. The world is divided into time zones of 15° (1 hour) each. There are ( ) time zones in total.
   A. 12 B. 24 C. 15

3. A place at 15° west longitude will see the dawn ( ) than a place at 15° east longitude.
   A. two hours later B. two hours earlier C. four hours later

[--Ans: 1. B; 2. B; 3. A--]

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SE4a_u1_09s_006_1

Look at the picture and think about the questions. Write the correct answers in the ( ).

Here is a piece of weather information from the Hong Kong Observatory: The weather is fine tonight and it will be cloudy tomorrow morning. There is northwest wind of force 2. The highest temperature tomorrow is 9°C and the lowest temperature is 3°C. The relative humidity is 50%. The air quality is good.



1. Which of the icons below represents the weather tomorrow? ( )

A.  B.  C. 

2. The weather forecast says the highest temperature tomorrow is 9°C. This temperature should appear tomorrow ( ).
   A. evening B. noon C. morning

3. The temperature difference between day and night tomorrow is about ( ) °C.
   A. 12 B. 9 C. 6

[--Ans: 1. A; 2. B; 3. C--]

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SE4a_u1_10s_001_1

Based on the experiment, answer the questions. Circle the answers or follow the instructions.

Fred observes the granite, sandstone and marble. He touches the rocks with his hands and observes with a magnifying glass.





1. When he touches granite and marble, which one is rougher?
   (granite / marble)

2. What can he observe from the rock samples with a magnifying glass?
   (size of particles / hardness)

3. Draw the particles of granite he observes in the box.

[--Ans: 1. granite; 2. size of particles; 3. There are different colours of particles, such as black and grey. The particle sizes are large. (Any reasonable answers.)--]

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SE4a_u1_10s_002_1

Based on the experiment and answer the questions. Circle the correct answers.



an experiment to measure the weight of the soil

1. The aim of experiment is to explore whether the soil contains a certain kind of component. What component is it?
   (air / water)

2. Before we measure the weight of the soil with a digital scale, what weight should we set to zero first?
   (the plate and soil on the digital scale / the plate on the digital scale)

3. A week later, what proves that there was water in the soil? (There can be more than one answer.)
   (The soil becomes much lighter. / There are animal and plant matter in the soil. / The soil becomes drier.)

[--Ans: 1. water; 2. the plate on the digital scale; 3. The soil becomes much lighter. The soil becomes drier.--]

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SE4a_u1_10s_003_1

Based on the experiments, answer the questions. Circle the correct answers.

Experiment 1



Experiment 2



1. In experiment 1, Jenny rubs some soil between her hands. She feels that the soil is wet. It shows that there is (water / air) in the soil.

2. In experiment 2, Fred puts some soil into water. He sees bubbles coming out. It shows that there is (water / air) in the soil.

3. Soil is a precious resource. After the experiments, the teacher and the pupils should put the soil into the (rubbish bin / flower bed).

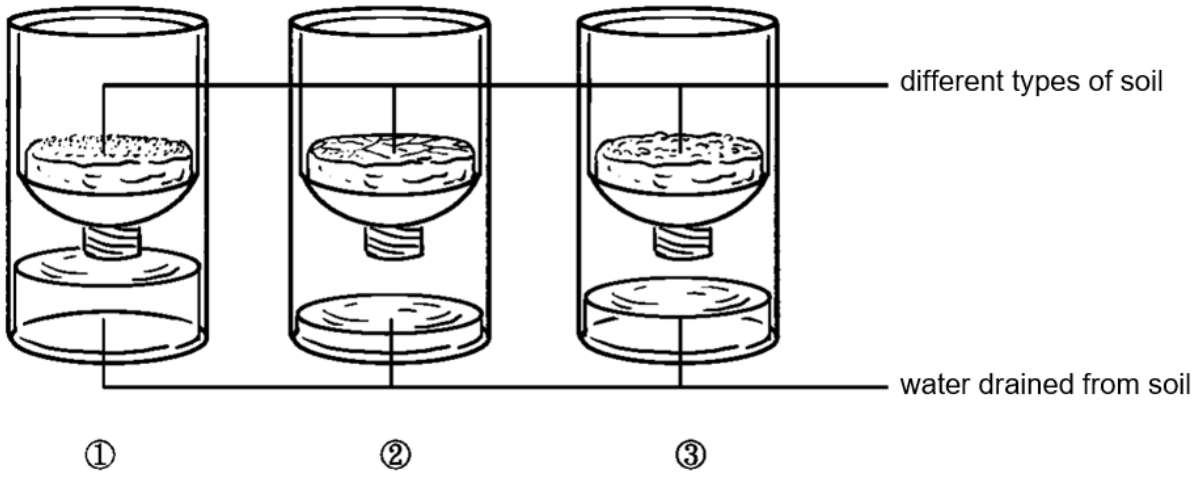
[--Ans: 1. water; 2. air; 3. flower bed--]

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SE4a_u1_10s_004_1

Based on the experiment, answer the questions. Circle the correct answers.

Jenny makes three water drainage bottles of the same size. She puts the same amount of dry soil into each bottle and pours the same amount of water into them. Then, she observes how well each type of soil drains.



1. What property of the three types of soil does this experiment compare?
   (drainage capacity / fertility)

2. From the results, we can tell that soil (① / ② / ③) has the best drainage capacity.
   The drainage capacity is the lowest in soil (① / ② / ③).

3. Refer to question 2. Which type of soil do you think sample ③ is?
   (sandy soil / loam / clay soil)

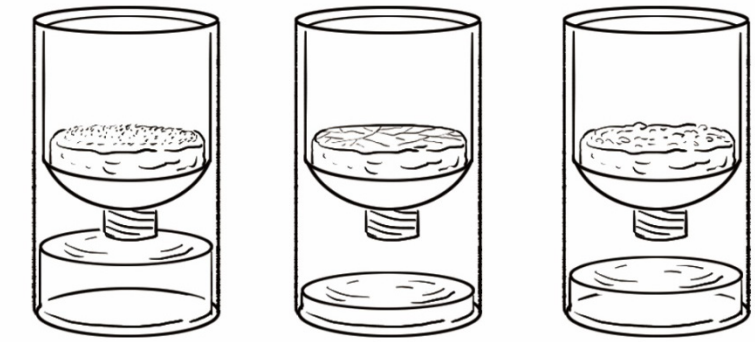
[--Ans: 1. drainage capacity; 2. ①, ②; 3. loam--]

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SE4a_u1_10s_005_1

Based on the experiment, answer the questions. Circle the correct answers.

Some pupils are studying the drainage capacity of different types of soil under the teacher's guidance. They use sandy soil, loam and clay soil to do the experiment.



1. The pupils observe and record the characteristics of the three types of soil. They find that they have different (particle sizes / compositions).

2. For fair test, the amount of the three types of soil samples should be (the same / different), and the amount of water poured should be (the same / different).

3. Five minutes later, the pupils pour the water that has been drained out into a measuring cup. They see that the amount of water from the bottle with sandy soil is the largest. This shows that the drainage capacity of sandy soil is the (worst / best).

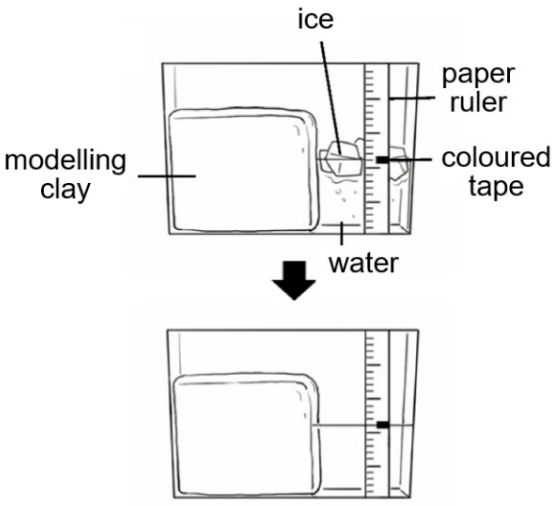
[--Ans: 1. particle sizes; 2. the same, the same; 3. best--]

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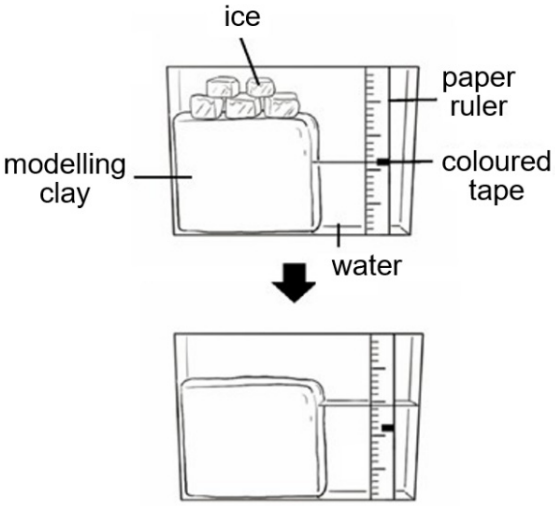
SE4a_u1_10s_007_1

Some pupils are exploring the effects of the melting of sea ice and land ice on sea levels. Based on the experiment, answer the questions. Write the answers on the ____ or circle the answers.

Experiment 1



Experiment 2



1. In the experiments, what do the ice on the modelling clay and the ice in the water represent?
   The ice on the modelling clay represents (sea ice / land ice). The ice in the water represents (sea ice / land ice).

2. What kind of experiment are these?
   (field experiment / simulation experiment)

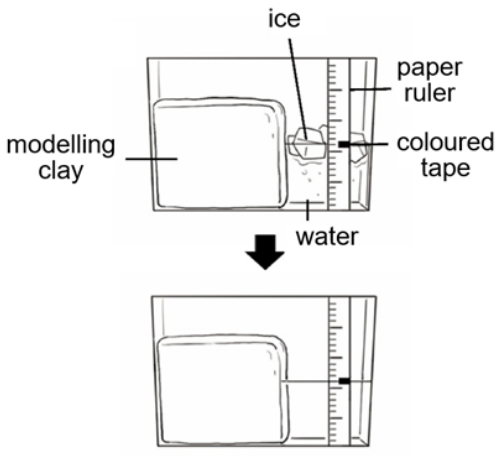
3. What is the use of marking the water level at the start with coloured tape?
   (It makes the ice melt faster. / It helps us observe the changes of the water level easily.)

[--Ans: 1. land ice, sea ice; 2. simulation experiment; 3. It helps us observe the changes of the water level easily.--]

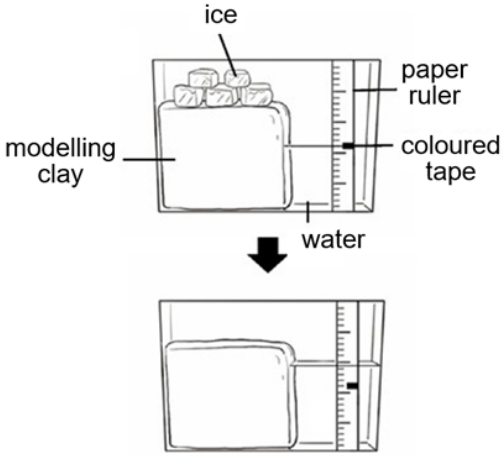
SE4a_u1_10s_008_1

Some pupils are studying the effects of melting sea ice and land ice on sea levels. Based on the experiments, answer the questions. Circle the correct answers.

Experiment 1



Experiment 2



1. Jenny sticks a paper ruler on the outside of each transparent box and puts (the same / different) amount of modelling clay and ice cubes inside the box.

2. Fred pours water into the boxes. There is enough water to make the ice cubes float, but without drowning the modelling clay. He measures the water level of each box and marks it on the side of the box with transparent coloured tape. This is the water level at the (start / end).

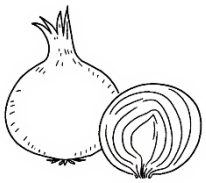
3. Jenny and Fred record the water levels until the ice cubes melt completely. In the box that simulate the melting of (land ice / sea ice), they find that the water level rises.

[--Ans: 1. the same; 2. start; 3. land ice--]

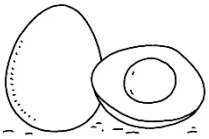
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SE4a_u1_12k_001_1

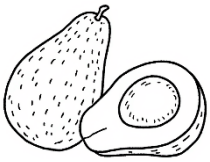
Which of these food items can be used to simulate the structure of the Earth?

A. 

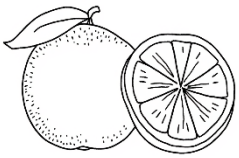
( )

B. 

( )

C. 

( )

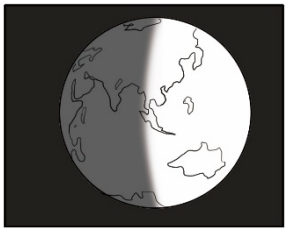
D. 

( )

[--Ans: A. ×; B. ✓; C. ✓; D. ×--]

SE4a_u1_12k_002_1

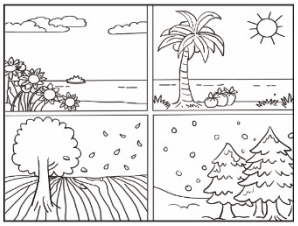
Which of these natural phenomena are caused by the movement and collision of plates?

A. 

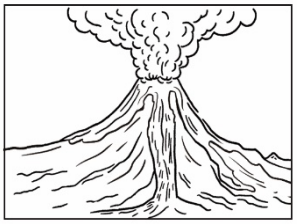
( )

B. 

( )

C. 

( )

D. 

( )

[--Ans: A. ×; B. ✓; C. ×; D. ✓--]

SE4a_u1_12k_003_1

Which of the following rocks are igneous rocks?

A. granite 

( )

B. sandstone 

( )

C. basalt 

( )

D. marble 

( )

[--Ans: A. ✓; B. ×; C. ✓; D. ×--]

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SE4a_u1_12k_004_1

Which of these items are made of rocks?

A. pebble path



( )

B. ceramic bowls



( )

C. stone mill



( )

D. stone pot



( )

[--Ans: A. ✓; B. ×; C. ✓; D. ✓--]

SE4a_u1_12k_005_1

Which of the following are the components of soil?

A. water



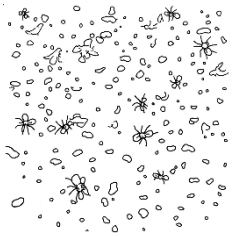
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B. plastic waste



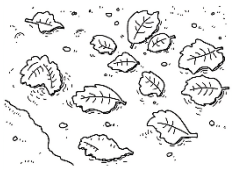
( )

C. animal matter



( )

D. plant matter



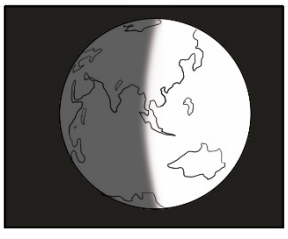
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[--Ans: A. ✓; B. ×; C. ✓; D. ✓--]

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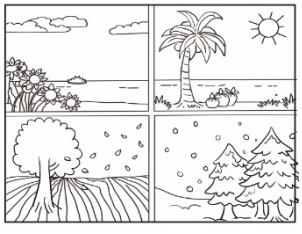
Which of these phenomena is caused by the rotation of the Earth?

A. day and night



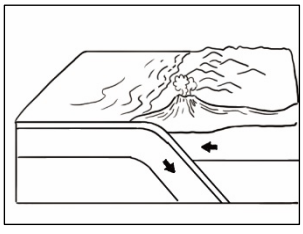
( )

B. changes of seasons



( )

C. movements of plates



( )

D. meteor



( )

[--Ans: A. ✓; B. ×; C. ×; D. ×--]

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SE4a_u1_12k_008_1

Which of the following are effects caused by global warming?

A.



( )

B.



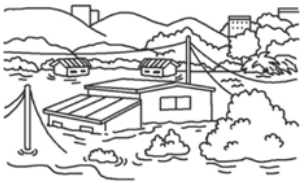
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C.



( )

D.



( )

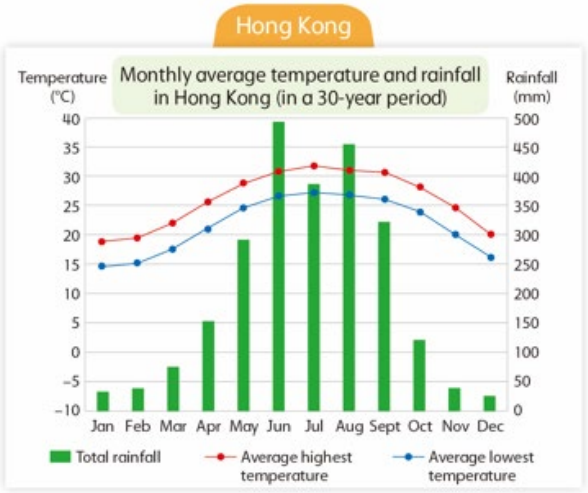
[--Ans: A. ✓; B. ×; C. ✓; D. ✓--]

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SE4a_u1_12s_001_1

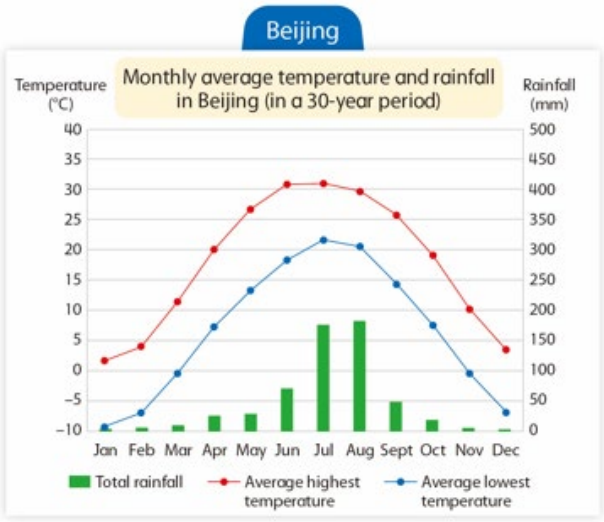
Which place has the smallest temperature changes over the year?

A. Hong Kong



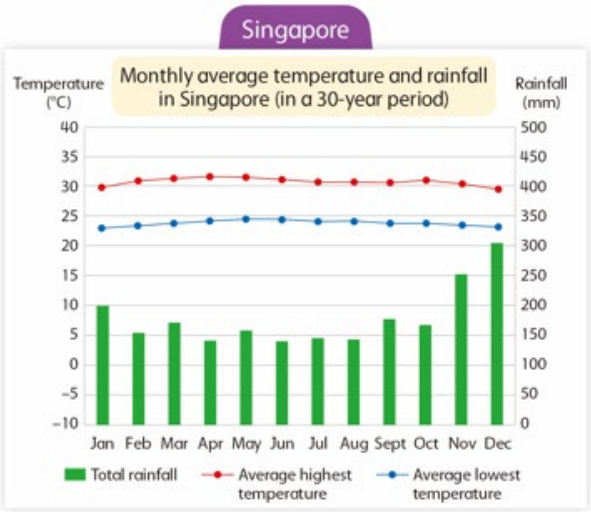
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B. Beijing



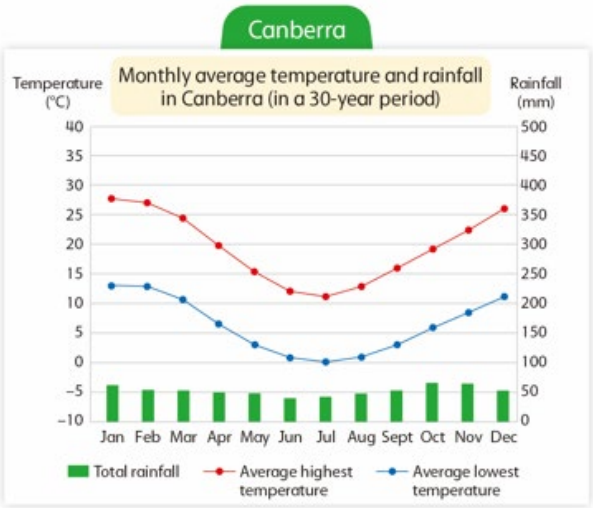
( )

C. Singapore



( )

D. Canberra



( )

[--Ans: A. ×; B. ×; C. ✓; D. ×--]

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SE4a_u1_12a_001_1

Which of these people increase their carbon footprints unnecessarily?

A. walk to school



( )

B. waste large amounts of food



( )

C. leave the electrical appliances on



( )

D. buy too much



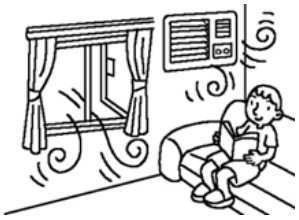
( )

[--Ans: A. ×; B. ✓; C. ✓; D. ✓--]

SE4a_u1_12a_002_1

Which of these people save energy in the daily life?

A.



( )

B.



( )

C.



( )

D.



( )

[--Ans: A. ×; B. ✓; C. ×; D. ✓--]

SE4a_u1_12a_003_1

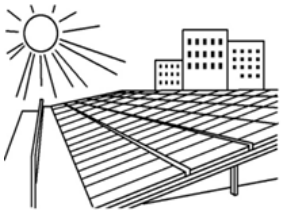
What should we do in the daily life to help slow down global warming?

A.



( )

B.



( )

C.



( )

D.



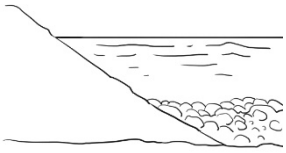
( )

[--Ans: A. ×; B. ✓; C. ✓; D. ✓--]

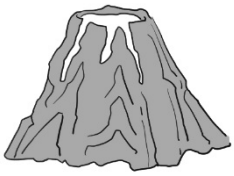
SE4a_u1_13k_001_1

Match the formation processes of rocks with the types of rocks.

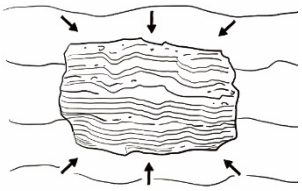
A.



B.



C.



1. igneous rocks ( )
2. sedimentary rocks ( )
3. metamorphic rocks ( )

[--Ans: 1. B; 2. A; 3. C--]

SE4a_u1_13k_002_1

Match the rocks with their types. (There can be more than one answer.)

A. shale



B. granite



C. sandstone



D. gneiss



1. igneous rocks ( )
2. sedimentary rocks ( )
3. metamorphic rocks ( )

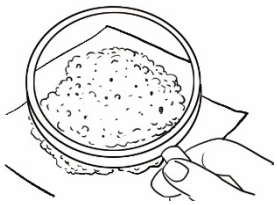
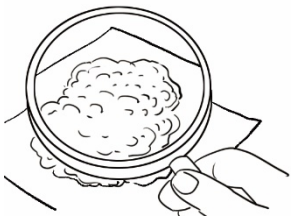
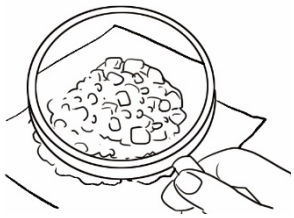
[--Ans: 1. B; 2. A, C; 3. D--]

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SE4a_u1_13k_003_1

Match the sizes of the soil particles with the types of soil.

A. B. C.

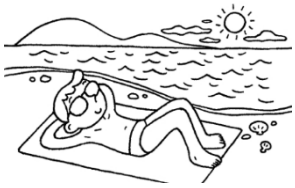
1. sandy soil ( )
2. loam ( )
3. clay soil ( )

[--Ans: 1. C; 2. B; 3. A--]

SE4a_u1_13k_004_1

When the northern hemisphere receives overhead sunlight, which of these are activities that are suitable to do in Hong Kong? Which of these are suitable to do in Canberra? (There can be more than one answer.)

A. B. C. D.



1. activities suitable to do in Hong Kong ( )
2. activities suitable to do in Canberra ( )

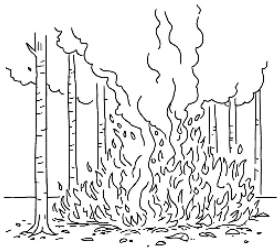
[--Ans: 1. B, C; 2. A, D--]

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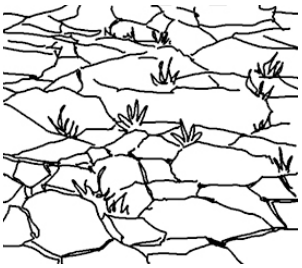
SE4a_u1_13k_006_1

Match the effects of global warming with the descriptions.

A. 

B. 

C. 

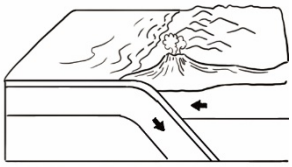
D. 

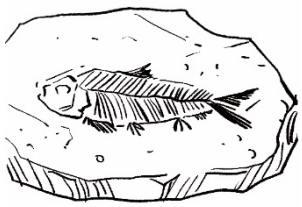
1. sandstorms ( )
2. melting of glaciers ( )
3. desertification ( )
4. wildfires ( )

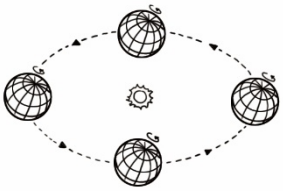
[--Ans: 1. A; 2. C; 3. D; 4. B--]

SE4a_u1_13k_007_1

Match the natural phenomena or human activities with their results.

A. 

B. 

C. 

D. 

1. changes of seasons ( )
2. volcanic eruption ( )
3. global warming ( )
4. formation of fossils ( )

[--Ans: 1. C; 2. A; 3. D; 4. B--]

SE4a_u1_13s_001_1

Scratch some rocks with fingernails and a stainless steel key. Match the results with the hardness of the rocks.

A. B. C. D.



1. least hard ( )
2. medium hardness ( )
3. hardest ( )
4. not sure ( )

[--Ans: 1. B; 2. D; 3. A; 4. C--]

SE4a_u1_13s_002_1

Match the methods of observing soil with their results.

A. B. C. D.



1. There is air in the soil. ( )
2. There is water in the soil. ( )
3. Soil is made up of particles of different grain sizes. ( )
4. There is animal and plant matter in the soil. ( )

[--Ans: 1. D; 2. B; 3. C; 4. A--]

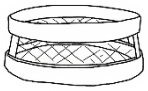
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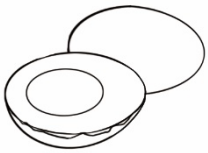
SE4a_u1_13s_003_1

What scientific knowledge can we gain from these items? Match the items with the descriptions.

A. 

B. 

C. 

D. 

1. the internal structure of the Earth ( )
2. that the Earth rotates ( )
3. how people in the past knew the time ( )
4. grain sizes of soil particles ( )

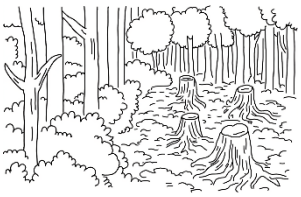
[--Ans: 1. D; 2. A; 3. B; 4. C--]

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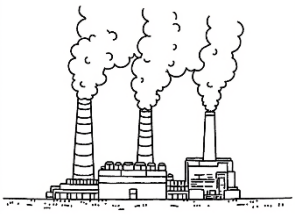
SE4a_u1_13a_001_1

Match the reasons for the increase in carbon dioxide in the atmosphere with the human activities.

A. 

B. 

C. 

D. 

1. burning rubbish releases large amounts of carbon dioxide ( )
2. the production, transportation and packaging of goods produce carbon footprints ( )
3. carbon dioxide cannot be fully absorbed by the forests due to deforestation ( )
4. factories give out large amounts of waste gas ( )

[--Ans: 1. A; 2. C; 3. B; 4. D--]

SE4a_u1_13a_002_2

Match the living habits with the reasons why they can reduce carbon footprints.

A. turn off the lights before going out 

B. recycle waste 

C. take public transport 

D. bring your own utensils 

1. it takes less energy to recycle than to make new materials ( )
2. it reduces carbon footprints during the production processes ( )
3. it reduces the waste gas from private cars ( )
4. it reduces the waste of electricity ( )

[--Ans: 1. B; 2. D; 3. C; 4. A--]

SE4a_u1_13a_003_1

Match the behaviours of reducing carbon footprints with the descriptions.

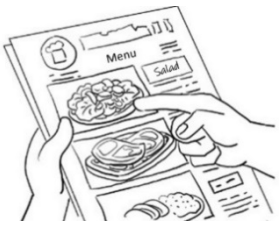
A.



B.



C.



D.



1. walk instead of taking public transport ( )
2. eat more vegetables and eat less meat ( )
3. turn off the lights before going out ( )
4. use less disposable utensils ( )

[--Ans: 1. B; 2. C; 3. D; 4. A--]

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SE4a_u1_15k_001_1

What is the centre of the Earth?

[--Ans: the core--]

SE4a_u1_15k_002_1

What is the outermost layer of the Earth?

[--Ans: the crust--]

SE4a_u1_15k_003_1

What is the outermost layer of the Earth mainly made up of?

[--Ans: rocks--]

SE4a_u1_15k_004_1

What gushes out of the volcano during a volcanic eruption?

[--Ans: magma--]

SE4a_u1_15k_005_1

What natural phenomena may plate movement and collision cause? Give one example.

[--Ans: earthquakes / volcanic eruptions (Any one.)--]

SE4a_u1_15k_008_1

What are some common igneous rocks? Give one example.

[--Ans: granite or basalt (Any one or other reasonable answers.)--]

SE4a_u1_15k_010_1

What are some common sedimentary rocks? Give one example.

[--Ans: sandstone or shale (Any one or other reasonable answers.)--]

SE4a_u1_15k_011_1

What are some common metamorphic rocks? Give one example.

[--Ans: marble or gneiss (Any one or other reasonable answers.)--]

SE4a_u1_15k_012_1

What characteristics of rocks can we observe?

[--Ans: their colours, grain sizes and hardness--]

SE4a_u1_15k_013_1

What are the main components of soil?

[--Ans: soil particles, animal and plant matter, water and air--]

SE4a_u1_15k_014_1

According to the grain size, which three types can soil be classified into?

[--Ans: sandy soil, loam, clay soil--]

SE4a_u1_15k_015_1

What does the Earth rotate on?

[--Ans: the axis--]

SE4a_u1_15k_016_1

How long does it take for the Earth to rotate once?

[--Ans: around 24 hours / 1 day--]

SE4a_u1_15k_018_1

The Earth moves around the Sun. What is this movement called?

[--Ans: revolution--]

SE4a_u1_15k_019_1

How long does it take for the Earth to complete one revolution?

[--Ans: 1 year / around 365 days--]

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SE4a_u1_15k_021_1

When the northern hemisphere receives slanted sunlight, what season is it in the northern hemisphere?

[--Ans: winter--]

SE4a_u1_15k_022_1

What are the effects of global warming? Give two examples.

[--Ans: melting of ice, rising sea levels, desertification, increase in natural disasters (Any two or other reasonable answers.)--]

SE4a_u1_15k_024_1

What gas should we reduce giving out to slow down global warming?

[--Ans: carbon dioxide / greenhouse gases--]

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SE4a_u1_15a_001_1

How can we protect rock and soil resources? Give two examples.

[--Ans: plant trees, do not litter, do not cut too many trees, promote knowledge about protecting natural resources (Any two or other reasonable answers.)--]

SE4a_u1_15a_002_1

Name two ways to reduce carbon footprints in the daily life.

[--Ans: do not buy too many things, save electricity, travel in public transport instead of private cars, do not waste food (Any two or other reasonable answers.)--]

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SE4a_u1_16k_001_2

Why do earthquakes and volcanic eruptions happen?

[--Ans: It is because the magma under the Earth's crust is moving constantly. The plates of the crust move and collide with each other, causing earthquakes or volcanic eruptions.--]

SE4a_u1_16k_003_2

According to what characteristics do we classify soil into sandy soil, loam and clay soil? What are the differences in the drainage capacity among these three types of soils?

[--Ans: We can classify soil according to the particle size. Sandy soil has the largest particles and the best drainage capacity; loam has medium-sized particles and medium drainage capacity; clay soil has the smallest particles and the worst drainage capacity.--]

SE4a_u1_16k_004_1

Why do we have day and night on the Earth?

[--Ans: The Earth rotates on its axis, which is an imaginary line that passes through the North Pole and the South Pole. The Earth keeps rotating, causing the cycle of day and night.--]

SE4a_u1_16k_005_2

Why are there seasonal changes on the Earth?

[--Ans: The Earth's axis is tilted. As the Earth revolves around the Sun, the angle of sunlight hitting different places on the Earth changes, causing seasonal changes.--]

SE4a_u1_16k_006_1

What is weather?

[--Ans: Weather is the condition of the atmosphere in a place over a short period of time.--]

SE4a_u1_16k_007_1

What is climate?

[--Ans: Climate is the average weather condition of a place over a longer period of time.--]

SE4a_u1_16k_010_1

What are carbon footprints?

[--Ans: Carbon footprints are the amount of greenhouse gases produced in some human activities.--]

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SE4a_u1_16a_001_2

As citizens, what can we do to protect rock and soil resources? Give one example.

[--Ans: join activities to clean the parks, rivers and nature reserves / choose to buy eco-friendly products that uses renewable sources to reduce the demand for natural resources (Any one or other reasonable answers.)--]

SE4a_u1_16a_002_2

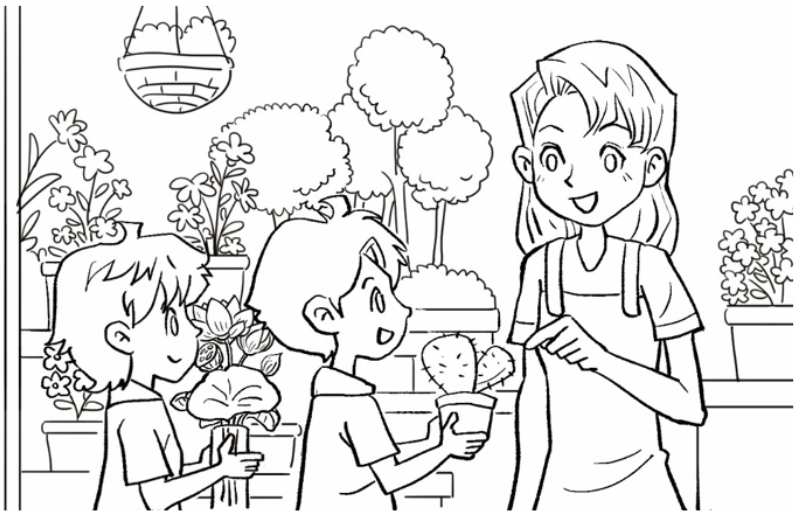
How can we live a life with low carbon footprints? Give two examples.

[--Ans: buy local food to reduce the carbon footprints during the transportation / eat less meat and more vegetables and grains to lower carbon footprints / take public transport such as the MTR and the bus and travel in private cars less (Any two or other reasonable answers.)--]

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SE4a_u1_17k_002_1

Fred and Jenny are choosing plants at the florist. Based on the situation, answer the questions. Circle the correct answers.



Jenny Fred

1. Cactuses can grow in dry places. Which type of soil is suitable for growing a cactus?
   (sandy soil / loam / clay soil)

2. Lotuses need fertile and sticky soil. Which type of soil is suitable for growing a lotus?
   (sandy soil / loam / clay soil)

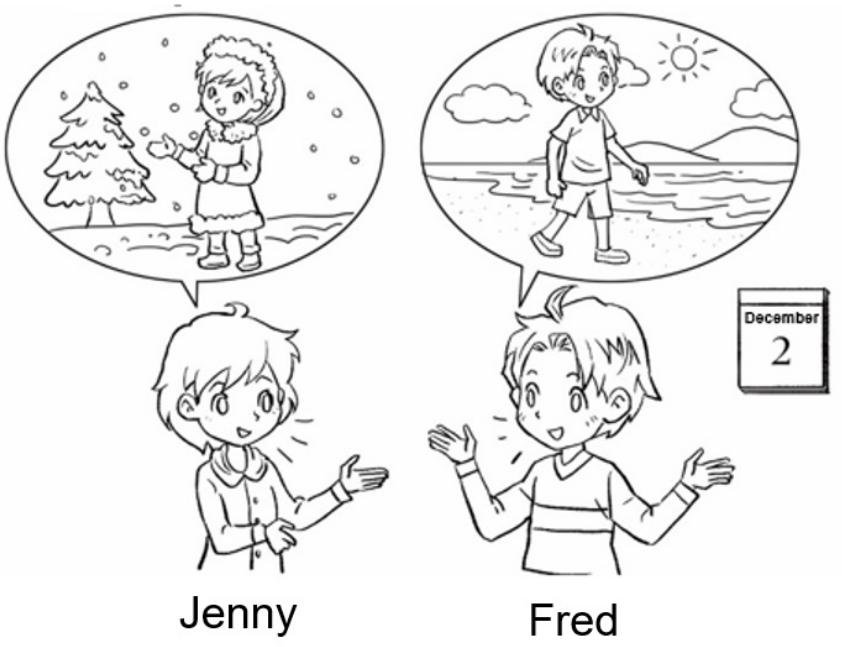
3. Jenny needs to get new soil for the roses at home. Which type of soil should she buy?
   (sandy soil / loam / clay soil)

[--Ans: 1. sandy soil; 2. clay soil; 3. loam--]

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SE4a_u1_17k_003_1

Jenny and Fred are talking about going on a trip with their family at Christmas. Based on the situation, answer the questions. Write the answers on the ____.



1. Based on the climate of Beijing and Canberra, which city should Fred and his family go?

   _____

2. Based on the climate of Beijing and Canberra, which city should Jenny and her family go?

   _____

3. Explain why you chose that city for Fred and his family.

   _____

[--Ans: 1. Canberra; 2. Beijing; 3. At Christmas, it is winter in the northern hemisphere and summer in the southern hemisphere. Therefore, Fred and his family should choose Canberra, which is in the southern hemisphere.--]

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SE4a_u1_17k_005_1

Jenny is checking the weather forecast to plan for the weekend. Based on the information, answer the questions. Write the correct letters in the ( ).

Weather Forecast for 6 June – 10 June				
Today (6 June)	Tomorrow (7 June)	Friday (8 June)	Saturday (9 June)	Sunday (10 June)
				
23 - 28°C	25 - 29°C	22 - 26°C	21 - 25°C	21 - 25°C
rainy	cloudy	sunny	rainy	cloudy
west wind	southwest wind	south wind	southeast wind	south wind

1. What aspects of the weather does the weather forecast in the table include? (There can be more than one answer.)
   A. temperature B. cloud amount C. precipitation
   D. wind direction E. wind speed

   ( )

2. Jenny plans to hang out on the weekend. According to the weather forecast, which of these pieces of advice is reasonable?
   A. She can go to outdoor playground on Saturday.
   B. She should bring an umbrella on Saturday.
   C. She should wear a jumper to stay warm on Sunday.

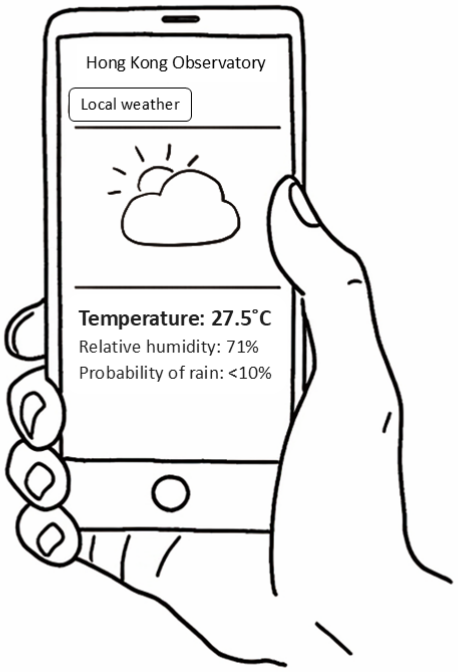
   ( )

[--Ans: 1. A, B, C, D; 2. B--]

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SE4a_u1_17k_006_1

Based on the situation, answer the questions. Write the answers on the ____.



1. What information does the weather forecast in the picture uses to describe the weather?

   _____

2. Besides measuring the weather on our own, from which organisation can we get weather information?

   _____

3. Weather changes every day. However, when we observe it over a long period of time, we can find a regular pattern for the changes. What is this pattern called?

   _____

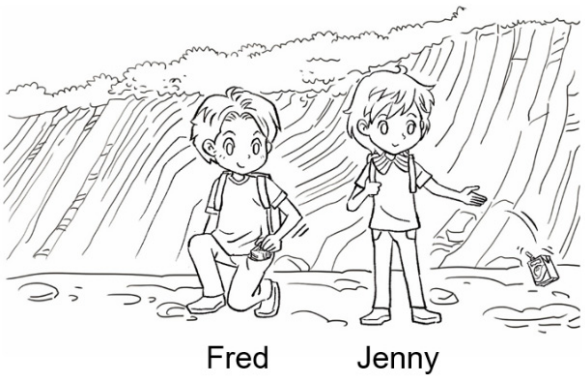
[--Ans: 1. Describe it with the temperature relative humidity and weather icon;
2. Hong Kong Observatory; 3. climate--]

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SE4a_u1_17a_001_1

Based on the situation, answer the questions. Write the correct letters in the ( ) or write the answer on the _____.

Fred and Jenny explore at Geopark.
They observe the rocks and soil there.



1. Fred picks up a rock to observe. What is he doing after that?
   A. taking the rock away
   B. putting back the rock
   C. throwing the rock to a far place

   ( )

2. Jenny drinks a box of drink. What is she doing after that?
   A. littering
   B. throwing the box into a rubbish bin
   C. taking away the box with her in her backpack

   ( )

3. What consequences may their actions lead to? (There can be more than one answer.)
   A. pollute the soil
   B. protect rock and soil resources
   C. damage the park

   ( )

4. Imagine you see them doing these. What will you tell them?

   _____

[--Ans: 1. A; 2. A; 3. A, C; 4. Rocks and soil are precious resources on the Earth. We should cherish and protect them. Do not damage the nature. (Any reasonable answers.)--]

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SE4a_u1_18k_001_2

Read the information and use what you have learned to answer the questions. Write the correct letters in the ( ) or write the answers on the ____.

The Earth is made of three layers: the crust, the mantle and the core. The crust is the land we stand on. It is made up of plates of different sizes.

The heat produced from the core makes the magma in the mantle flow. The magma rises from the lower mantle and cools when it reaches the upper mantle. The magma spreads out and then sinks back down to the lower mantle. This causes convection and plate movement. As the strength of the convection is different in different parts of the mantle, different plates move at different speeds and in different directions. As a result, the plates rub against and collide with one another. Every time that happens, energy builds up. When the crust cannot hold the energy anymore, huge amounts of energy are released, causing the plates to move vigorously. The vigorous movement or breaking of the edges of the plates releases seismic waves, causing the ground to shake and resulting in natural hazards like earthquakes.

1. Put the formation processes of the convection inside the Earth in the correct order.
   A. the magma rises from the lower mantle to the upper mantle
   B. the core produces heat
   C. the magma sinks back down to the lower mantle
   D. the magma cools and spreads out
   ( ) → ( ) → ( ) → ( )

2. How are earthquakes formed?
   _____
   _____

3. Name one natural hazard that earthquakes may cause.
   _____

[--Ans: 1. B → A → D → C; 2. The plates rub against and collide with one another. Every time that happens, energy builds up. When the crust cannot hold the energy anymore, huge amounts of energy are released, causing the plates to move vigorously; 3. volcanic eruptions, tsunamis, landslide, avalanche (Any one or other reasonable answers.)--]

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SE4a_u1_18k_002_1

Read the information and use what you have learned to answer the questions. Write the answers on the ____ or write the correct letters in the ( ).

Did you know how hot the inside of the Earth is? According to scientists' estimation, the temperature of the inner core is around 4400°C – 6000°C, which is similar to the temperature of the Sun's surface. The Earth was formed 4.5 billion years ago. Why has the temperature of the inner core not decreased over time?

Some radioactive substances inside the Earth (such as uranium-238, thorium-232 and radium-226) produce a large amount of heat energy during their natural decay process. They provide a continuous supply of heat to the inside of the Earth. In addition, the thick crust blocks the heat from escaping, just like a vacuum bottle keeping hot water warm. Under the combined effects of heating and heat preservation, the inside of the Earth still has a very high temperature after 4.5 billion years.

1. The temperature of the core is between _____ and _____.

2. Why is the temperature of the core high? (There can be more than one answer.)
   A. the thick crust helps keep the Earth warm
   B. living things on the Earth provide energy for the Earth
   C. the Sun provides energy for the Earth
   D. radioactive substances inside the Earth provide heat for the Earth

( )

[--Ans: 1. 4400°C, 6000°C; 2. A, D--]

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SE4a_u1_18k_003_2

Read the information and use what you have learned to answer the questions. Write the correct letters in the ( ).

When large rocks are exposed to the effects of sunlight, wind, water, earthquakes, and volcanoes over a long period of time, they gradually begin to break apart. This process is called weathering.

Through weathering caused by sunlight, rain, and other natural forces, rocks slowly break down into smaller rock fragments. These fragments are carried by wind, water, and other natural forces to low-lying areas, eventually forming loose soil. Plants begin to grow, and microorganisms break down plant remains into organic matter and minerals in the soil, making it fertile. The process of soil formation is slow, but soil can disappear and degrade very quickly.

1. Rocks become soil through ( ).
   A. dissolving B. weathering C. burning

2. Which of these breaks down plant remains into organic matter and minerals in the soil?
   A. animals
   B. plants
   C. sunlight
   D. microorganisms

   ( )

3. According to the information, what components are there in soil?
   A. rock fragments
   B. organic matter
   C. minerals
   D. all of the above

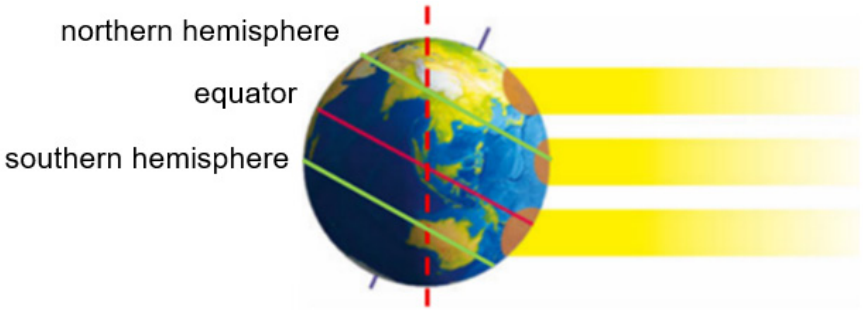
   ( )

[--Ans: 1. B; 2. D; 3. D--]

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SE4a_u1_18k_004_1

Read the information and use what you have learned to answer the questions. Circle the answers.



The equator divides the Earth into the northern hemisphere and the southern hemisphere. The northern hemisphere is the area north of the equator, while the southern hemisphere is the area south of the equator. The North Pole is the northernmost point of the northern hemisphere, while the South Pole is the southernmost point of the southern hemisphere.

As the axis is tilted, sunlight hits the Earth's surface at different angles in different places on different days. When the northern hemisphere receives overhead sunlight, it is summer in the northern hemisphere. The daytime is longer than the night-time. When the southern hemisphere receives overhead sunlight, it is winter in the northern hemisphere. The daytime is shorter than the night-time. Except for places on the equator, the length of day and night changes throughout the year in other places.

1. Why are the seasons in the northern and southern hemispheres opposite?
   The Earth's axis is (upright / tilted) so the amount of sunlight the northern and southern hemispheres receive is (the same / different).

2. According to the picture, the area that receives sunlight in the northern hemisphere is (larger / smaller) so the daytime is (longer / shorter) while the night-time is (longer / shorter).

3. There is no change in the length of day and night in places (in the northern hemisphere / on the equator / in the southern hemisphere).

[--Ans: 1. tilted, different; 2. larger, longer, shorter; 3. on the equator--]

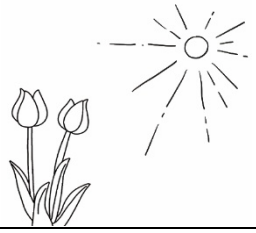
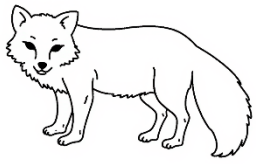
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SE4a_u1_18k_005_1

Read the information and use what you have learned to answer the questions. Write the answers on the ____.

Light has the greatest impact on living things during the day and night. The light is stronger during the day and weaker at night. Some animals adapt to stronger light and mainly move in the daytime. They are called diurnal animals, such as butterflies. Some animals adapt to weaker light and mainly move at night. They are called nocturnal animals, such as owls and bats. Tulips open their petals as the Sun rises and close them at night. Other plants that change with the day and night, like tulips, include the night-blooming cactus and evening primrose.

The changes of the four seasons also affect living things. Some animals change their fur in different seasons. For example, the arctic fox living in the North Pole changes its fur to different colours in summer and winter. Some animals, such as brown bears and hedgehogs, go into a deep sleep to get through the cold winter. Some plants, such as maple trees and poplar trees, shed their leaves in autumn and winter.


1. The factor that affects living things the most during the day and night is _____.

2. Some animals adapt to bright light and are active during the day. They are called _____. Some animals adapt to weaker light and are active at night. They are called _____.

3. How do the changing seasons affect living things? Give one example.
_____
_____

[--Ans: 1. light; 2. diurnal animals, nocturnal animals; 3. Arctic foxes have different colours of fur in summer and winter / brown bears go into a deep sleep in winter / some plants shed their leaves in autumn and winter (Any one or other reasonable answers.)--]

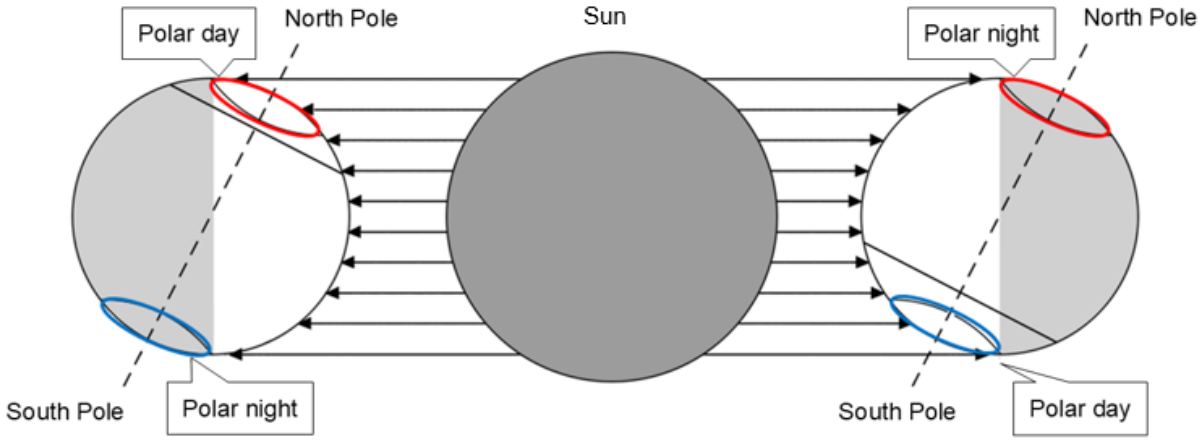
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SE4a_u1_18k_006_2

Read the information and use what you have learned to answer the questions. Write the answers on the ____.

Polar day and polar night are natural phenomena unique to the polar regions. Polar day is when the Sun never sets and the sky is always bright. Polar night is when the Sun never rises and the sky is always dark. Polar day and polar night occur because the Earth revolves around the Sun in an oval-shaped orbit while rotating around its tilted axis.

Polar day and polar night mainly occur in areas north of the Arctic Circle and south of the Antarctic Circle. When the Sun shines directly on the northern hemisphere, polar day occurs north of the Arctic Circle, while polar night occurs south of the Antarctic Circle. When the Sun shines directly on the southern hemisphere, polar night occurs north of the Arctic Circle, while polar day occurs south of the Antarctic Circle. The closer you get to the poles, the longer the periods of polar day and polar night become, lasting up to six months.



1. When the Earth revolves around the Sun, it also rotates around its tilted axis. What unique natural phenomena occur in the polar regions?

_____

2. What is polar day?

_____

3. What is polar night?

_____

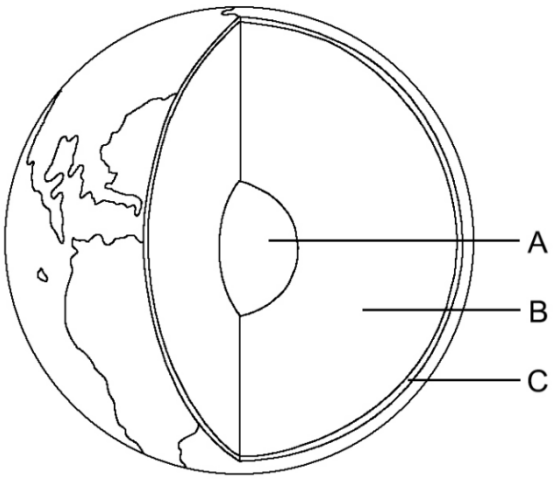
_____

4. When it is summer in the northern hemisphere, there is _____ in the area north of the Arctic Circle and _____ in the area south of the Antarctic Circle.

[--Ans: 1. polar day and polar night; 2. It is when the Sun does not set and the sky is always bright in the polar regions of the Earth.; 3. It is when the Sun does not rise and the sky is always dark in the polar regions of the Earth.; 4. polar day, polar night--]

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What is the structure of the Earth? Write the correct names on the _____.

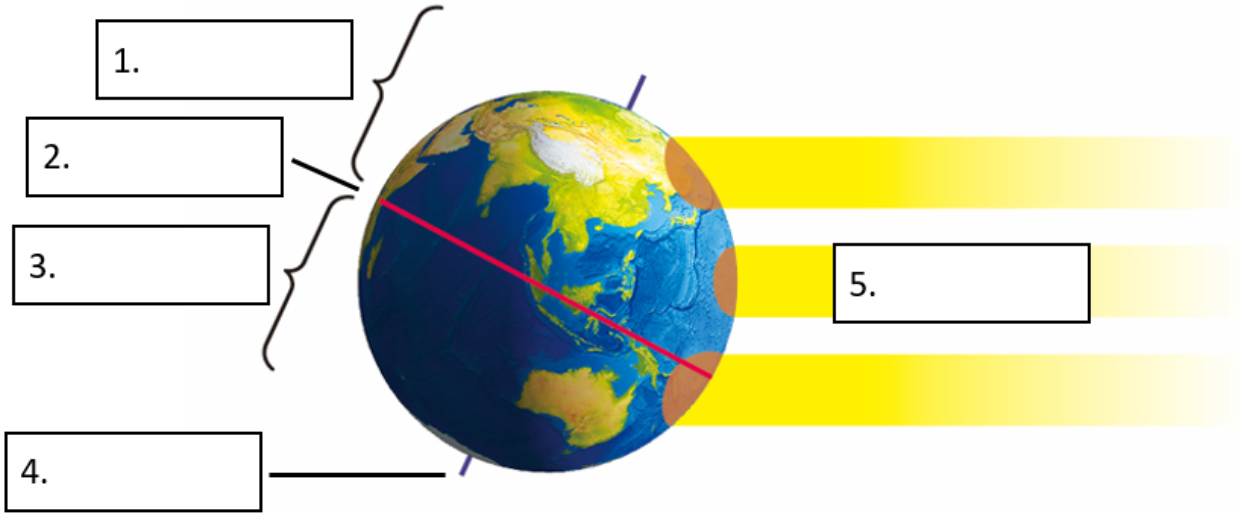


A. _____ B. _____ C. _____

[--Ans: A. core; B. mantle; C. crust--]

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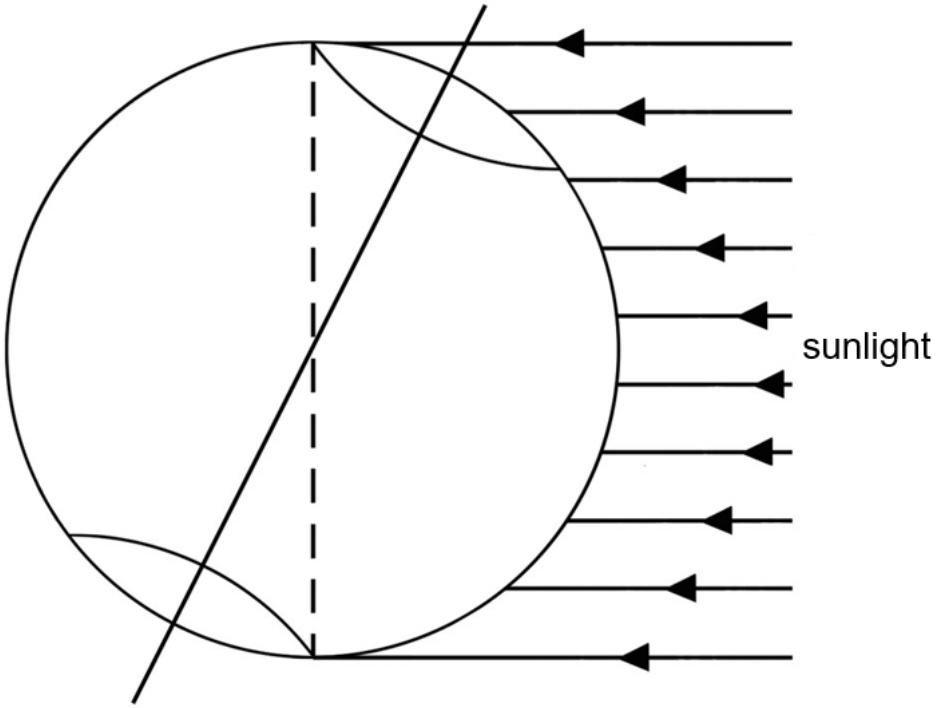
Based on the picture, write the correct names in the boxes.



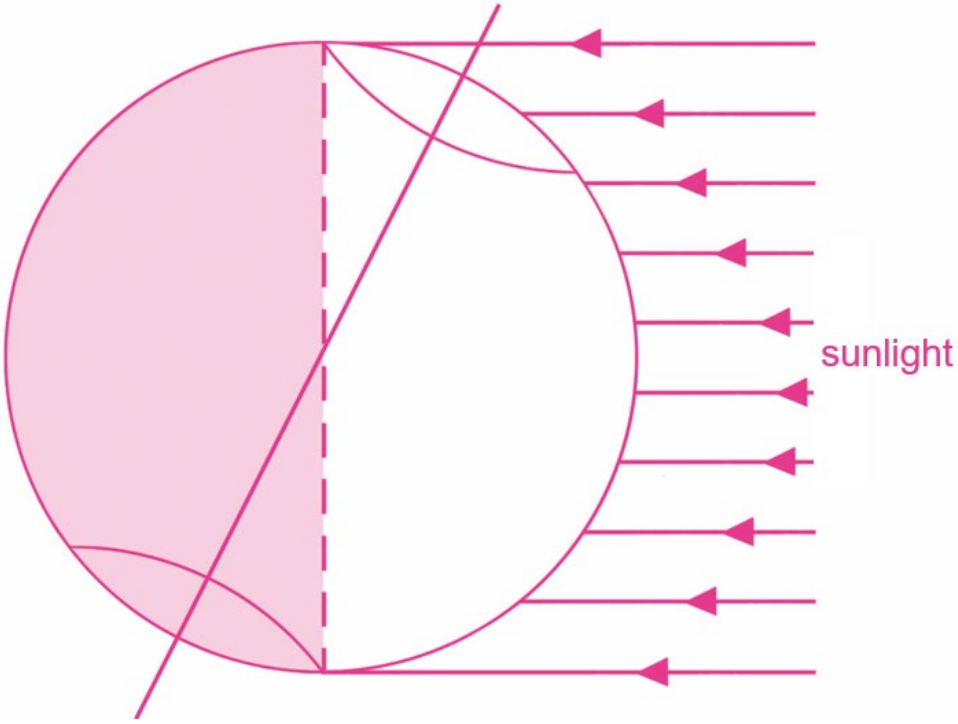
[--Ans: 1. northern hemisphere; 2. equator; 3. southern hemisphere; 4. Earth's axis; 5. sunlight--]

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Colour the area that do not receive sunlight in grey.



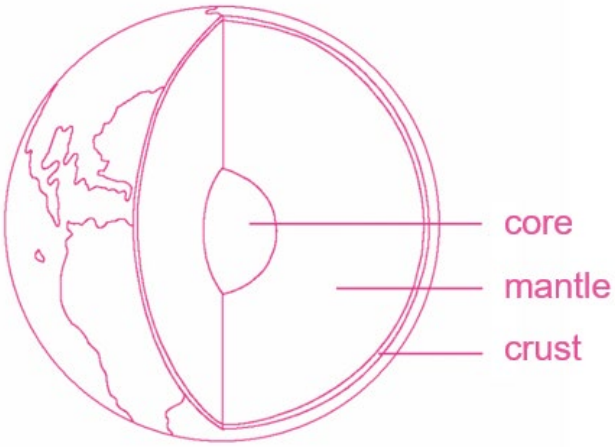
[--Ans: see below--]



SE4a_u1_21k_001_1

Draw the structure of the Earth and label the layers.

[--Ans: see below--]



SE4a_u1_21a_001_1

Draw one way in the daily life to reduce carbon footprints.

☺ Reduce carbon footprints in the daily life ☺

We can do this! ✌

[--Ans: walk instead of travelling in cars / bring your own shopping bag / do not waste food / do not waste electricity / do not use disposable utensils (Any reasonable answers.)--]